

A GOLDEN RATIO-BASED CONSTANT AND CONJECTURES ON ITS POLYLOGARITHMIC VALUES

TRISTEN HARR

ABSTRACT. This paper introduces and analyzes a specific complex constant, denoted Λ_{G1} , which is uniquely derived from a set of defining equations related to the golden ratio, ϕ . The motivation for this constant is shown to arise from a classical geometric construction. We first prove the constant's fundamental properties, including the convergence condition $|\Lambda_{G1}| < 1$. The primary contribution of this work is the systematic identification of the values of two families of infinite series where Λ_{G1} serves as the argument, connecting them to Eulerian polynomials and the Polylogarithm function, $\text{Li}_s(\Lambda_{G1})$. Furthermore, we present high-precision numerical evaluations for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) of this constant and, based on this evidence, we conjecture that these values are transcendental numbers not contained within the field extension $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.

1. INTRODUCTION

The study of special functions and constants often reveals deep connections between different areas of mathematics. This paper focuses on the properties of a specific complex constant derived from the golden ratio, ϕ . The motivation for this constant arises from the classical geometric problem of dividing a line segment in extreme and mean ratio. This problem provides a non-arbitrary foundation for a pair of defining equations whose unique solution gives rise to the constant we analyze.

The purpose of this paper is to rigorously define this constant, Λ_{G1} , and systematically explore the evaluation of several important families of infinite series using it as the primary argument. We will show that because this constant is proven to lie within the unit disk, it serves as a valid argument for well-known summation formulas. This work provides a careful cataloging of these results and extends the analysis to a numerical investigation of the resulting values of the Polylogarithm function, culminating in a formal conjecture about their nature. The constant's constrained algebraic structure makes it a compelling candidate for such a study.

2. THE CONSTANT Λ_{G1} AND ITS FOUNDATIONAL PROPERTIES

2.1. Defining Equations and Uniqueness. We define two real constants, T and J , as the unique solution to a pair of equations motivated by a classical geometric problem.

Definition 2.1 (Defining Equations). The constants T and J are the real numbers (a, b) that satisfy:

- (1) $a + b = 1/2$.
- (2) $a/b = \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$.

Solving this system yields the unique values $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$.

From these constants, we define our primary object of study.

Definition 2.2 (The Primary Constant). The constant Λ_{G1} is defined as the complex number $\Lambda_{G1} = T + iJ$.

2.2. Convergence Property. A crucial property of Λ_{G1} is that its magnitude is less than one, which guarantees the convergence of the power series we will investigate.

Lemma 2.3 (Magnitude of Λ_{G1}). *The constant Λ_{G1} lies within the unit disk, i.e., $|\Lambda_{G1}| < 1$.*

Proof. Convergence requires $|\Lambda_{G1}|^2 < 1$. We calculate the squared values of the components: $T^2 = \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16}$ and $J^2 = \left(\frac{3-\sqrt{5}}{4}\right)^2 = \frac{14-6\sqrt{5}}{16}$. Summing these gives:

$$\begin{aligned} |\Lambda_{G1}|^2 &= T^2 + J^2 = \frac{6 - 2\sqrt{5} + 14 - 6\sqrt{5}}{16} \\ &= \frac{20 - 8\sqrt{5}}{16} = \frac{5 - 2\sqrt{5}}{4} \approx 0.131966 \end{aligned}$$

Since $0.131966 < 1$, the condition is satisfied. $\square$

3. APPLICATION TO POLYNOMIAL-WEIGHTED SERIES

The first family of series we identify are those weighted by polynomial powers of the index. As established in Lemma 2.3, $|\Lambda_{G1}| < 1$, which allows for its direct substitution into the standard formula for such series. The sum is given in terms of the Eulerian polynomials, $A_k(x)$ [1]. For instance, the first two Eulerian polynomials are $A_0(x) = 1$ and $A_1(x) = x$. The general identification for our constant is:

$$\sum_{n=0}^{\infty} n^k \cdot (\Lambda_{G1})^n = \frac{A_k(\Lambda_{G1})}{(1 - \Lambda_{G1})^{k+1}}$$

This identity holds for any integer $k \geq 0$.

4. APPLICATION TO RECIPROCAL-WEIGHTED SERIES

The second family of series are those weighted by reciprocal powers of the index. As with the polynomial case, the convergence of Λ_{G1} allows for its direct use in the defining series for the Polylogarithm function [2].

Definition 4.1 (The Polylogarithm Function). The Polylogarithm function of order s , $\text{Li}_s(z)$, is defined by the power series:

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

This series converges for all complex numbers z such that $|z| < 1$.

Since $|\Lambda_{G1}| < 1$, the series $\sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s}$ converges, and its value is, by definition, the Polylogarithm of Λ_{G1} :

$$\sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s} = \text{Li}_s(\Lambda_{G1})$$

This identification holds for any complex s .

5. NUMERICAL ANALYSIS AND A CONJECTURE ON THE NATURE OF $\text{Li}_s(\Lambda_{G1})$

We now investigate the nature of the specific values identified in the previous section.

5.1. Numerical Values. We present high-precision numerical evaluations for the first two non-trivial integer cases of the Polylogarithm of Λ_{G1} . The values were computed using the series definition in Python with the ‘mpmath’ library, at a working precision of 100 decimal places.

- For $s = 2$, the Dilogarithm of Λ_{G1} is:

$$\text{Li}_2(\Lambda_{G1}) \approx 0.322328253720993 + 0.226730769307749i$$

- For $s = 3$, the Trilogarithm of Λ_{G1} is:

$$\text{Li}_3(\Lambda_{G1}) \approx 0.310345112613931 + 0.198884393433621i$$

5.2. A Conjecture on the Nature of These Values. The transcendental nature of special function values at specific algebraic arguments is a deep and active area of research in number theory [2]. Our numerical results motivate a new conjecture in this domain. Analysis using integer relation algorithms (such as PSLQ) fails to resolve the values above into simple, closed-form expressions involving common mathematical constants. This suggests a tension between the algebraic nature of the argument Λ_{G1} and the analytic nature of the Li_s function. We therefore propose the following.

Conjecture 5.1. *For any integer $s \geq 2$, the value $\text{Li}_s(\Lambda_{G1})$ is a transcendental number. Furthermore, it is conjectured that this number is not in the field $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.*

Remark. Proving this conjecture is beyond the scope of this paper, but proposing it frames a clear direction for future work and elevates the significance of the constant Λ_{G1} .

6. CONCLUSION

We have rigorously defined a complex constant, Λ_{G1} , derived from equations motivated by a classical geometric problem. We have proven its convergence, which validates its use in a variety of infinite series. The primary contribution of this work is the systematic identification of the values for series weighted by n^k and n^{-s} evaluated at this constant.

Furthermore, we have extended this analysis by providing high-precision numerical values for $\text{Li}_2(\Lambda_{G1})$ and $\text{Li}_3(\Lambda_{G1})$. Based on this evidence, we have conjectured that these values are transcendental numbers not constructible from common constants. This paper thus provides a focused, self-contained exploration, a set of benchmark results, and a new open problem for the community.

7. FUTURE WORK

The conjecture presented in Section 5 opens a clear path for future investigation. A primary goal would be to prove or disprove the transcendental nature of $\text{Li}_s(\Lambda_{G1})$. Further work could also explore other families of series or integral transforms evaluated at Λ_{G1} , or investigate whether the constant's unique algebraic properties have applications in other areas of mathematics or physics, such as dynamical systems or crystallography.

APPENDIX A. GEOMETRIC MOTIVATION FOR THE DEFINING EQUATIONS

The defining equations in Section 2 are motivated by a classical compass-and-straightedge construction. Specifically, they represent the algebraic solution to the problem of dividing a line segment of length 1/2 into two smaller segments, T and J , such that their ratio is the golden ratio, i.e., $T/J = \phi$. This geometric origin provides a foundational, non-arbitrary basis for the constants.

REFERENCES

- [1] M. Abramowitz and I. A. Stegun, eds., *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Dover Publications, New York, 1965.
- [2] L. Lewin, *Polylogarithms and Associated Functions*. North Holland, New York, 1981.